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A Report on Case study

" Shortest Path Algorithms in Google Maps and Navigation Systems"

by

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II B.Tech, I Semester

COMPUTER SCIENCE AND ENGINEERING(AI & ML)

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(AUTONOMOUS), ANANTHAPURAMU

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Title of the Course:	Advanced Data Structures and Algorithm Analysis
Course Code:	23CSE301
Class, Semester, Section:	II B. Tech, I Semester
Course Type:	Core
Regulation:	SRIT R23
Course Coordinator:	K.Rubiya Alam
Title of the Case study:	Shortest Path Algorithms in Google Maps and Navigation Systems
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1. Introduction

Shortest Path Algorithms are important algorithms in Data Structures and Algorithms (DSA). They are used to find a path between two points with minimum cost.

In Google Maps and navigation systems, roads and locations are represented as a **weighted graph**. Locations are represented as **vertices**, while roads are represented as **edges**. The edge weight may represent distance, travel time, traffic delay, fuel cost, or toll cost.

Therefore, the shortest path does not always mean the physically shortest road. A slightly longer road may be faster when another road has heavy traffic.

2.Background

Navigation systems need to find suitable routes between different locations. A user may have several possible roads to reach a destination.

Shortest-path algorithms help the system select routes based on:

- Distance
- Travel time
- Traffic

Two important algorithms used for shortest-path problems are **Dijkstra's Algorithm** and **A* Algorithm**.

Dijkstra's Algorithm

Dijkstra's algorithm finds shortest paths from a source vertex to other vertices when the edge weights are non-negative.

- Select the starting location.

Set its distance to 0.

A*Algorithm

A* is another shortest-path algorithm. It uses the cost already travelled and an estimated cost to the destination.

Formula: f(n) = g(n) + h(n)

g(n) = actual cost from the starting point to node n; h(n) = estimated cost from node n to the destination; f(n) = estimated total cost.

3. Case study overview

Google Maps is a digital mapping and navigation application that provides directions between different locations. For example, if a student wants to travel from **Home to College**, there may be several possible roads. The navigation system represents the road network as a graph and searches for a suitable route.

Objectives

- To find a suitable route between two locations.
- To reduce travel distance and time.
- To consider traffic and road conditions.
- To support real-time navigation and rerouting.

Scope

- Google Maps and navigation systems
- Public transportation
- Emergency services
- Robotics

4.Methodology

Graph Representation: Locations or intersections are represented as vertices, and roads are represented as edges. Distance or travel time is used as the edge weight. The starting location is the source and the destination is the target.

Dijkstra's Algorithm: The system selects the source, initializes distances, repeatedly selects the closest unvisited location, checks its neighbours, updates distances, and continues until the destination is reached.

A* Algorithm: A* uses the actual cost and estimated remaining cost: $f(n) = g(n) + h(n)$.

Traffic Consideration: A 5 km route with heavy traffic may take 30 minutes, while a 7 km route with low traffic may take 18 minutes. Therefore, travel time can be used as an edge weight.

Route Recalculation: If the original route becomes congested or unavailable, system can calculate another route. Road weights can change according to traffic and road connections.

5. Results & Benefits

Shortest-path algorithms allow navigation systems to automatically find suitable routes.

Example: Route 1: A -> B -> E = 5 + 7 = 12

Route 2: A -> C -> D -> E = 2 + 1 + 2 + 3 = 8

Therefore, for the given example, A -> C ->D -> E has the smaller total cost.

Benefits

- Finds routes automatically.
- Saves travel time.
- Reduces unnecessary distance.
- Handles large road networks.
- Provides alternative routes.
- Responds to changing road conditions.

6. Challenges & Solutions

Challenge	Solution
Changing traffic	Use dynamic edge weights and recalculate the route.
Large road networks	Use suitable algorithms such as A*.
One-way roads	Represent them using directed graphs.
Road closures	Remove the road or assign a very high cost.
GPS errors	Combine location information with the road network.
Multiple objectives	Consider distance, time, traffic, tolls and user preferences.

7. Conclusion

Shortest Path Algorithms help Google Maps find suitable routes between locations. Algorithms like *Dijkstra's and A** consider distance, travel time, traffic, and road conditions. They also support rerouting when conditions change, making navigation faster and more efficient.

8. Evaluation Parameters

Criteria	Marks	Marks Obtained
Problem Understanding & Clarity	**2**	
Use of Aids (Figures/Charts/Models)	**2**	
Application of Course Concepts	**2**	
Methodology & Analysis	**2**	
Results, Findings & Discussion	**2**	

9. Faculty Feedback / Remarks

The case study report on Shortest Path Algorithms in Google Maps and Navigation Systems is well-prepared and demonstrates a clear understanding of algorithmic applications in real-world navigation systems. The introduction and background sections effectively explain the importance of shortest path algorithms in route planning. The methodology clearly presents Dijkstra's and A* algorithms and their use in finding suitable routes.

The results section provides practical insights into route optimization, traffic consideration, and route recalculation. The inclusion of challenges and solutions shows an understanding of practical limitations such as changing traffic, road closures, GPS errors, and large road networks. The report is coherent, well-organized, and written in a professional academic style.

References

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C., Introduction to Algorithms, MIT Press.
2. Ahuja, R. K., Magnanti, T. L., & Orlin, J. B., Network Flows: Theory, Algorithms, and Applications, Prentice Hall.

Signature of the Student